VMC Vidyamandir Classes SINCE 1986
IIT JEE | MEDICAL | FOUNDATION

CODE
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# JEE ADVANCED - 11 | PAPER - 2

## JEE 2022

| Date | Maximam Marks | Duration |
|---|---|---|
| 27th March, 2022 | 180 | 3 Hours |

## General Instructions

1. The test is of **3 hours** duration and the maximum marks are **180.**
2. The question paper consists of 3 Subject (Subject I: **Physics**, Subject II: **Chemistry**, Subject III : **Mathematics**). Each Part has **three** sections (Section 1, Section 2 & Section 3).
3. **Section 1** contains **SIX (06) Multiple Correct Answers Type Questions**. Each question has 4 choices (A), (B), (C) and (D), out of which **ONE OR MORE THAN ONE CHOICE** is correct.
4. **Section 2** contains **EIGHT (08) Numerical Value Type Questions**. The answer to each question is a **NUMERICAL VALUE**. For each question, enter the correct numerical value of the answer. If the answer is a decimal numerical value, then round-off the value to TWO decimal places. If the answer is an Integer value, then do not add zero in the decimal places. *In the OMR, do not bubble the ⊕ sign for positive values. However, for negative values, Θ sign should be bubbled*. (Example: 6, 81, 1.50, 3.25, 0.08)
5. **Section 3** contains **FOUR (04)** questions. Each question has **TWO (02)** matching lists: **LIST-I** and **LIST-II**. **FOUR** options are given representing matching of elements from **LIST-I** and **LIST-II**. **ONLY ONE** of these four options corresponds to a correct matching. For each question, choose the option corresponding to the correct matching.
6. For answering a question, an ANSWER SHEET (OMR SHEET) is provided separately. Please fill your **Test Code, Roll No.** and **Group** properly in the space given in the ANSWER SHEET.

Name of the Candidate (In CAPITALS) : ______________________

Roll Number : ______________________

OMR Bar Code Number : ______________________

Candidate's Signature : ______________ Invigilator's Signature ______________

## MARKING SCHEME

### SECTION-I

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
- For each question, choose the correct option(s) to answer the question.
- Answer to each question will be evaluated according to the following marking scheme:

| | | | |
|---|---|---|---|
| Full Marks | : | **+4** | If only (all) the correct option(s) is (are) chosen. |
| Partial Marks | : | **+3** | If all the four options are correct but ONLY three options are chosen. |
| Partial Marks | : | **+2** | If three or more options are correct but ONLY two options are chosen, both of which are correct options. |
| Partial Marks | : | **+1** | If two or more options are correct but ONLY one option is chosen and it is a correct option. |
| Zero Marks | : | **0** | If none of the options is chosen (i.e. the question is unanswered). |
| Negative Marks | : | **–2** | In all other cases. |

- **For Example:** If first, third and fourth are the ONLY three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

### SECTION-II

- This section has **EIGHT (08)** Questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer. If the answer is a decimal numerical value, then round-off the value to TWO decimal places. If the answer is an Integer value, then do not add zero in the decimal places. *In the OMR, do not bubble 15the ⊕ sign for positive values. However, for negative values, Θ sign should be bubbled*. (Example: 6, 81, 1.50, 3.25, 0.08)
- Answer to each question will be evaluated according to the following marking scheme:

**Full Marks:** +3 If ONLY the correct numerical value is entered.

**Zero Marks:** 0 In all other cases.

### SECTION-III

- This section contains **FOUR (04)** questions.
- Each question has **TWO (02)** matching lists: **LIST-I** and **LIST-II**.
- **FOUR** options are given representing matching of elements from **LIST-I** and **LIST-II**. **ONLY ONE** of these four options corresponds to a correct matching.
- For each question, choose the option corresponding to the correct matching.
- For each question, marks will be awarded according to the following marking scheme:

| | | | |
|---|---|---|---|
| Full Marks | : | **+3** | If ONLY the option corresponding to the correct matching is chosen. |
| Zero Marks | : | **0** | If none of the options is chosen (i.e. the question is unanswered). |
| Negative Marks | : | **–1** | In all other cases. |

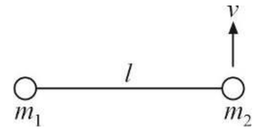
**SUBJECT I : PHYSICS** **60 MARKS**

## SECTION 1

## MULTIPLE CORRECT ANSWERS TYPE

**This section consists of SIX (06) Multiple Correct Answers Type Questions**. Each question has 4 choices (A), (B), (C) and (D), out of which **ONE OR MORE THAN ONE CHOICE** is correct.

**1.** Two small particles $m_1$ and $m_2$ being connected by an inextensible string of length $l$ are kept on a smooth horizontal surface. The particle $m_2$ is projected with a velocity $v$ perpendicular to the string. Choose the correct option(s).




**(A)** Angular velocity of $m_2$ w.r.t. $m_1$ is $\dfrac{m_2 v}{(m_1+m_2)\ell}$

**(B)** After time $t=\dfrac{\pi\ell}{2v}$ string will be perpendicular to its initial orientation

**(C)** Tension in the string in subsequent motion is $\dfrac{m_2 v^2}{\ell}$

**(D)** Displacement of $m_1$ in time $\dfrac{\pi\ell}{v}$ is $\dfrac{m_2\ell}{m_1+m_2}\sqrt{\pi^2+4}$

**2.** Consider a disk of mass $m$, radius $R$ lying on a liquid layer of thickness $T$ and coefficient of viscosity $\eta$. Which of the following is correct ?

**(A)** For the disk to rotate at a constant angular velocity $\omega$, we need to apply a tangential force $\dfrac{\eta\, 2\pi\omega R^3}{3T}$ at a distance of $\dfrac{3R}{4}$ from the centre

**(B)** The power required to rotate the disk at a constant angular velocity $\omega$ is given by $\dfrac{\pi\omega^2\eta R^4}{2T}$

**(C)** The power required to rotate the disk at a constant angular velocity $\omega$ is given by $\dfrac{2\pi\omega^2\eta R^4}{T}$

**(D)** For the disk to rotate at a constant angular velocity $\omega$, we need to apply a tangential force $\dfrac{\eta\,\pi\omega R^3}{3T}$ at a distance of $\dfrac{3R}{4}$ from the centre

**3.** Consider a long narrow cylinder of cross-section $A$ filled with a compressible liquid upto a height $h$ whose density $\rho$ is a function of pressure $P(z)$ *as* $\rho(z)=\dfrac{\rho_0}{2}\left(1+\dfrac{P(z)}{P_0}\right)$, where $P_0$ and $\rho_0$ are constants. The depth $z$ is measured from the free surface of the liquid where pressure is equal to atmospheric pressure $(P_{\text{atm}})$. Choose the correct options.

**(A)** Density varies with depth as $\rho(z)=\dfrac{\rho_0(P_0+P_{\text{atm}})}{2P_0}\, e^{\rho_0 gz/2P_0}$

**(B)** Mass of the liquid in container is $\dfrac{A}{g}(P_0+P_{\text{atm}})\,(e^{\rho_0 gh/2P_0}-1)$

**(C)** Pressure varies with depth as $P(z)=P_{\text{atm}}e^{\rho_0 gz/2P_0}-P_0$

**(D)** $\dfrac{dP(z)}{dz}=\dfrac{\rho_0}{2}\left(1+\dfrac{P(z)}{P_0}\right)g$

**4.** Taking into account the motion of the nucleus of a hydrogen atom, choose the correct statements. Take $\mu = \frac{M\, m_e}{m_e + M}$; $M$ : Mass of proton, $\hbar = h/2\pi$; $h$: planks constant

**(A)** Binding energy of electron in ground state is $\frac{\mu e^4}{32\pi^2 \in_0^2 \hbar^2}$

**(B)** Radius of nth orbit is $\frac{4\pi \in_0 n^2 \hbar^2}{\mu e^2}$

**(C)** Binding energy if motion of nuclei is not taken into account is more than the binding energy if nucleus motion is considered

**(D)** If $\frac{m_e}{M} = 0.00055$, percentage by which B.E. (Nucleus motion taken into account) differs from B.E. (Nucleus at rest) is 0.055%.

**5.** An opaque sphere of radius $R$ lies on a horizontal plane. On the perpendicular through the point of contact, there is a point source of light at a distance $R$ above the top of the sphere (i.e., $3R$ from the plane)

**(A)** Shadow of the sphere on the plane will be a circle

**(B)** Area of shadow on the plane will be $3\pi R^2$

**(C)** Area of the shadow on the plane is $2\pi R^2$

**(D)** A transparent liquid of refractive index $\sqrt{3}$ is filled above the plane such that the sphere is just covered with liquid. Area of shadow of sphere on plane will now be $2\pi R^2$

**6.** In a radioactive decay chain, ${}^{238}_{92}\mathrm{U}$ nucleus decays to ${}^{206}_{82}\mathrm{Pb}$ nucleus. Let $a$ and $b$ be the number of $\alpha$ and $\beta^-$ particles, respectively, emitted in this decay process. Which of the following statements is (are) true ?

**(A)** $a = 8$ **(B)** $a = 6$ **(C)** $b = 2$ **(D)** $b = 6$

## SECTION 2

## NUMERICAL VALUE TYPE QUESTIONS

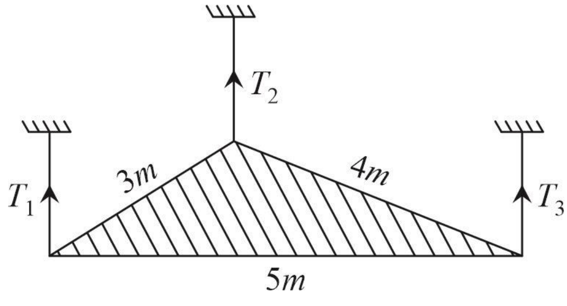
**This section consists of EIGHT (08) Numerical Value Type Questions**. The answer to each question is a **NUMERICAL VALUE**. For each question, enter the correct numerical value of the answer. If the answer is a decimal numerical value, then round-off the value to TWO decimal places. If the answer is an Integer value, then do not add zero in the decimal places. *In the OMR, do not bubble the ⊕ sign for positive values. However, for negative values, Θ sign should be bubbled*. (Example: 6, 81, 1.50, 3.25, 0.08).

**7.** A metal plate is exposed to light with wavelength 5000 *nm*. It is observed that electrons are ejected from the surface of the plate. When a retarding uniform electric field 0.01 *N*/*C*. is imposed, no electron can move away from the plate farther than a certain distance *d*. If the threshold wavelength is twice the incident light wavelength for the material of the plate then the distance *d* is _______*m*.
(Use $hc = 12400\,eV\text{Å}$)

**8.** A ball is thrown from the foot of a plane whose inclination to horizontal is 30°. It strikes the inclined plane at a distance 0.7 *m* from the point of projection. After collision with the inclined plane it rebounds, without any loss of energy, and follows its former path exactly in reverse. The speed with which it was projected is ________ *m*/*s*.

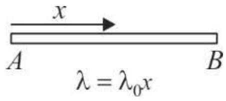
**9.** Two heating coils A and B are used to heat water. When coil A is supplied a constant voltage *V*, it heats 1 kg water from 20°C to 60°C in 10 minutes. When coil B is supplied a constant voltage *V*, it heats 1 kg water from 20°C to 80°C in 5 minutes. Now, the coils are connected in series and the combination is supplied a constant voltage *V*. The temperature (starting from 20°C) to which the coils heat 1 kg of water in 15 minutes is ________°C.
(Assume that water is heated in insulated vessels such that the heat loss to the surroundings is negligible)

**10.** A uniform triangular plate of mass $m = 4$ kg is tied at the three vertices by threads keeping the plate in equilibrium and such that the plane of the plate is kept is horizontal as shown. Side lengths of the plate are as shown. What is the value of $2T_1 + T_2 - T_3$ (in Newton)? Take [ $g = 10\text{ m/s}^2$ ]




**11.** A square loop of side $a = 0.5\,m$ with sides parallel to $x$ and $y$ axis is moved with velocity $v = 2\,m/s$ in the positive *x*-direction in a magnetic field along the positive *z*-direction. The field is neither uniform in space nor constant in time. It has a gradient $\frac{dB}{dx} = -10^{-3}\frac{T}{m}$ along the *x*-direction and it is increasing with time at rate $\frac{dB}{dt} = 10^{-3}\frac{T}{s}$. What is magnitude of current (in Ampere) in the loop if its resistance is $0.1\,m\Omega$?

**12.** A thin rod $AB$ of length $l$ has non uniform mass distribution such that its mass per unit length varies with distance $x$ from end A as $\lambda = \lambda_0 x$, where $\lambda_0$ is a positive constant of appropriate dimensions. If it is hinged at end $A$ and a force $F$ is applied normal to the rod at end $B$, the angular acceleration of the rod is $\alpha_A$. However if it is hinged at end $B$ and the force $F$ is applied normal to the rod at end A, the angular acceleration of the rod is $\alpha_B$. Find $\frac{\alpha_B}{\alpha_A}$. (Assume gravity free space)




**13.** A satellite of mass $2m$ is in a circular orbit of radius $2R$ around the earth. By mistake another satellite of mass $m$ is put in the same orbit and having opposite sense of rotation. The collision between the two satellites is perfectly inelastic and combined mass strikes the earth's surface at an angle $\theta$ with horizontal. Value of $\cos\theta$ is $\frac{2}{\sqrt{n}}$, where $n$ is _______.

**14.** $\vec{A}$ is a constant vector of magnitude 20 and $\vec{B}$ has magnitude 10 but its direction is unknown. Angle between $\vec{A}$ and $\vec{B}$ so that $\vec{A}+\vec{B}$ makes maximum angle with $\vec{A}$ is $10n$ in degree. Value of $n$ is________.

## SECTION 3

## MATRIX MATCH TYPE

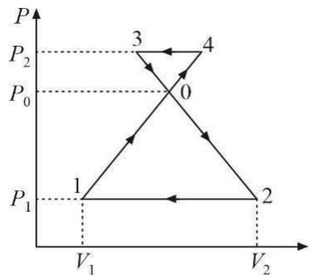
This section contains **FOUR (04)** questions. Each question has **TWO (02)** matching lists: **LIST-I** and **LIST-II**. **FOUR** options are given representing matching of elements from **LIST-I** and **LIST-II**. **ONLY ONE** of these four options corresponds to a correct matching. For each question, choose the option corresponding to the correct matching.

**15.** The magnetic field $B$ is measured at a point $P(0, 0, d)$ generated due to various current distributions and the dependence of $B$ on $d$ is found to be different for different current distributions. List-I contains different relations between $B$ and $d$. List-II describes different current distributions, along with their locations. Match the functions in List-I with the related current distributions in List-II.

| **List-I** | | **List-II** | |
|---|---|---|---|
| **(P)** | $B$ is independent of $d$ | **(1)** | A long hollow cylindrical current carrying wire, of radius $l$, with its axis coincident with the y-axis. Take $2l << d$ |
| **(Q)** | $B \propto \frac{1}{d}$ | **(2)** | A small current carrying ring in the $xy$ plane with centre at origin and radius $l$. Take $2l << d$ |
| **(R)** | $B \propto \frac{1}{d^2}$ | **(3)** | An infinite current carrying wire coincident with $x$-axis |
| **(S)** | $B \propto \frac{1}{d^3}$ | **(4)** | Two infinite current carrying wires parallel to the $x$-axis. The one along ($y$ = 0, $z$ = $l$) has a current $I$ and the one along ($y$ = 0, $z$ = $-l$) has a current $-I$. Take $2l << d$ |
| | | **(5)** | Infinite plane coincident with the $xy$ plane with uniform current per unit length $k$ along $x$ |

**(A)** (P)-(5), (Q)-(3, 4), (R)-(1), (S)-(2)
**(B)** (P)-(5), (Q)-(1, 3), (R)-(4), (S)-(2)
**(C)** (P)-(5), (Q)-(3), (R)-(1, 2), (S)-(4)
**(D)** (P)-(4), (Q)-(2, 3), (R)-(1), (S)-(5)

**16.** One mole of a monoatomic ideal gas undergoes thermodynamics processes as shown schematically in the $PV$-diagram below. Process 1-4 is a straight line passing through origin. Match the statements in List-I with the corresponding statements in List-II.




| List-I | | List-II | |
|---|---|---|---|
| **(P)** | Work done by the gas is zero | **(1)** | In process 2-1 |
| **(Q)** | Temperature of the gas remains unchanged | **(2)** | In process 4-3 |
| **(R)** | Work done is positive | **(3)** | In process 0-4 |
| **(S)** | Work done is negative | **(4)** | In process 0-4-3-0 |
| | | **(5)** | In process 0-2-1-0 |
| | | **(6)** | In process 1-0 |
| | | **(7)** | None of these |

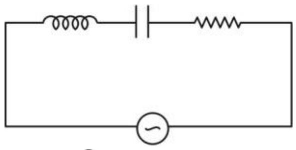
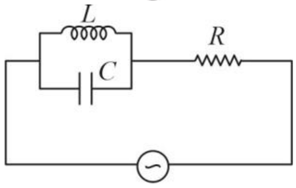
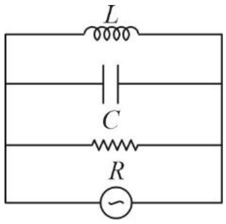
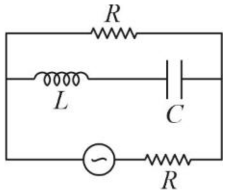
**(A)** (P)-(7), (Q)-(4, 5), (R)-(3, 5, 6), (S)-(1, 2, 4)

**(B)** (P)-(7), (Q)-(3, 5, 6), (R)-(2, 5), (S)-(4)

**(C)** (P)-(3, 5), (Q)-(4, 6), (R)-(1), (S)-(2)

**(D)** (P)-(7), (Q)-(4, 6), (R)-(2), (S)-(1)

**17.** Let $X_L, X_C$ be the inductive reactance and capacitive reactance and $R$ be the resistance in each of the circuits given in List-I. Let $V_L, V_C$ and $V_R$ be the r.m.s. voltage drop in each case and $i$ be the r.m.s. current from main and $I_1$ is the r.m.s current passing through $R$. The supply voltage is $10\,V$, $X_L = 10\,\Omega$, $X_C = 10\,\Omega$ and $R = 10\,\Omega$. Match the entries in List-I to all the entries in List-II.

| List-I | | List-II | |
|---|---|---|---|
| **(P)** |  | **(1)** | $i = 1\ A$ |
| **(Q)** |  | **(2)** | $i = 0$ |
| **(R)** |  | **(3)** | $V_L = 10\,V$ |
| **(S)** |  | **(4)** | $V_C = 10\ V$ |
| | | **(5)** | $I_1 = 1A$ |

**(A)** (P)-(1, 3, 4, 5), (Q)-(2, 3, 4), (R)-(1, 4, 5), (S)-(1, 3, 4, 5)
**(B)** (P)-(1, 5), (Q)-(2, 3, 4), (R)-(1, 3, 5), (S)-(1, 3, 5)
**(C)** (P)-(1, 3, 4, 5), (Q)-(2, 3, 4), (R)-(1, 3, 4, 5), (S)-(1, 3, 4, 5)
**(D)** (P)-(1, 3, 4), (Q)-(2, 3, 4), (R)-(1, 3, 4, 5), (S)-(1, 3, 4, 5)

**18.** In the List-I below, four different paths of a particle are given as functions of time. In these functions, $\alpha$ and $\beta$ are positive constants of appropriate dimensions and $\alpha \neq \beta$. In each case, the force acting on the particle is either zero or conservative. In List-II, certain statements are given about speed and angular velocity of the particle about origin. Match each path in List-I with those quantities in List-II.

| **List-I** | | **List-II** | |
|---|---|---|---|
| **(P)** | $\vec{r}(t) = \alpha t\,\hat{i} + \beta t\,\hat{j}$ | **(1)** | Speed is constant |
| **(Q)** | $\vec{r}(t) = \alpha(\cos\omega t\,\hat{i} + \sin\omega t\,\hat{j})$ | **(2)** | Speed is decreasing with time |
| **(R)** | $\vec{r}(t) = \alpha\,\hat{i} + \beta t\,\hat{j}$ | **(3)** | Speed is increasing with time |
| **(S)** | $\vec{r}(t) = \alpha t\,\hat{i} + \frac{\beta}{2} t^2\,\hat{j}$ | **(4)** | Angular velocity is constant |
| | | **(5)** | Angular velocity is decreasing with time |
| | | **(6)** | Angular velocity is increasing with time |

**(A)** (P)-(1, 5), (Q)-(2, 5), (R)-(3, 4), (S)-(5)
**(B)** (P)-(1, 4), (Q)-(1, 4), (R)-(1, 5), (S)-(3, 5)
**(C)** (P)-(2, 4), (Q)-(3, 5), (R)-(1, 4), (S)-(2, 5)
**(D)** (P)-(3, 5), (Q)-(2, 5), (R)-(2, 4), (S)-(2, 5)

## SUBJECT II : CHEMISTRY 60 MARKS

## SECTION 1

## MULTIPLE CORRECT ANSWERS TYPE

**This section consists of SIX (06) Multiple Correct Answers Type Questions**. Each question has 4 choices (A), (B), (C) and (D), out of which **ONE OR MORE THAN ONE CHOICE** is correct.

**1.** Which of the following statement(s) is/are true about the complex ion $[Pt(en)_2Cl_2]^{2+}$? $[en = H_2N\,CH_2\,CH_2\,NH_2]$

**(A)** It has two geometrical isomers-cis and trans

**(B)** Both the cis and trans isomers display optical activity

**(C)** Only the cis isomer has non-superimposable mirror image

**(D)** The oxidation state of Pt in the complex is +4

**2.** Which of the following statement(s) is/are true?

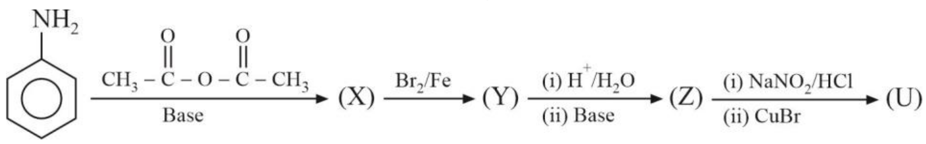
**(A)** $Ag^{+}$ ions give white precipitate with excess of concentrated HCl

**(B)** $Cu^{2+}$ ions produce a white precipitate when KCN solution is added in a small quantity and allowed to stand

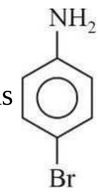
**(C)** $Hg^{2+}$ ions give deep blue precipitate with cobalt acetate and ammonium thiocyanate

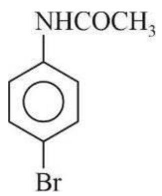
**(D)** Black precipitate of $BiI_3$ turns white when heated with water

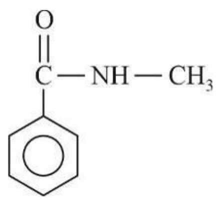
**3.** The correct statement(s) for the following reaction sequence is/are :



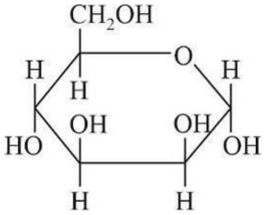

**(A)** The compound (U) is 

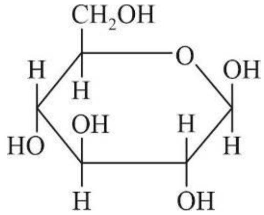

**(B)** The compound (Z) is 

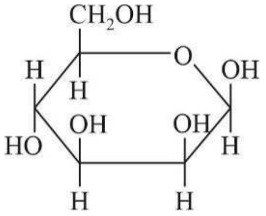

**(C)** The compound (Y) is 

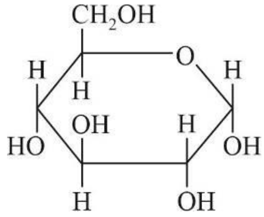

**(D)** The compound (X) is 

**4.** D-Mannose differs from D-glucose in its stereochemistry at C – 2. Which is/are not the correct structure of pyranose form of D-Mannose.

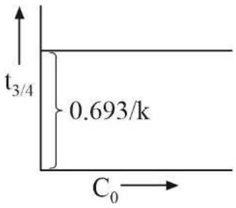
**(A)** 

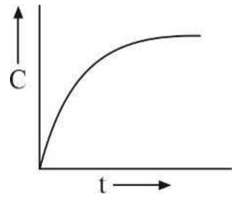
**(B)** 

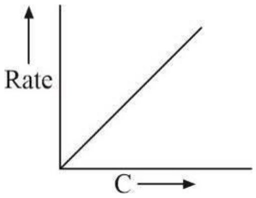
**(C)** 

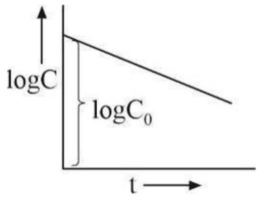
**(D)** 

**5.** For the first order reaction $A_{(g)} \longrightarrow B_{(g)} + C_{(g)}$ at constant volume and 300 K the concentration of A at time t = 0 is $C_0$ and the concentration of A after time t = t is C. $t_{3/4}$ represents the time required to reach the concentration of A to $C_0/4$. The correct option(s) is/are: (Given k is rate constant)

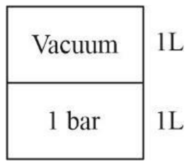
**(A)** 


**(B)** 


**(C)** 


**(D)** 


**6.** A container of volume 2L is separated into equal compartments. In one ompartment one mole of an ideal monoatomic gas is filled at 1 bar pressure and the other compartment is completely evacuated. A pinhole is made in the separator so gas expands to occupy full 2L and heat is supplied to gas so that finally pressure of gas equals 1 bar. The correct statement(s) is/are : (Given : 1 litre bar = 100J)




**(A)** $\Delta U = 150$ J

**(B)** $\Delta H = 250$ J

**(C)** Entropy change of system is greater than zero

**(D)** Work done by gas is zero

## SECTION 2

## NUMERICAL VALUE TYPE QUESTIONS

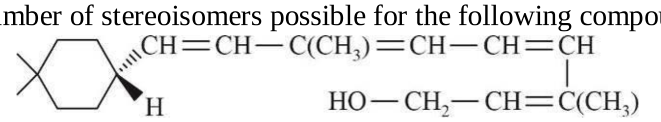
**This section consists of EIGHT (08) Numerical Value Type Questions**. The answer to each question is a **NUMERICAL VALUE**. For each question, enter the correct numerical value of the answer. If the answer is a decimal numerical value, then round-off the value to TWO decimal places. If the answer is an Integer value, then do not add zero in the decimal places. *In the OMR, do not bubble the ⊕ sign for positive values. However, for negative values, Θ sign should be bubbled*. (Example: 6, 81, 1.50, 3.25, 0.08).

**7.** The total number of compound having $p\pi - d\pi$ bonding among the molecules given below are $SO_2$, $SO_3$, $CO_2$, $H_3PO_4$, $H_3BO_3$, $HClO_4$, $HNO_3$, $HOCl$

**8.** The mass of Zn (atomic mass = 65) required to recover Ag from a 500 ml solution of 0.5 M sodium argento cyanide is x gram. The value of 16x is _______.

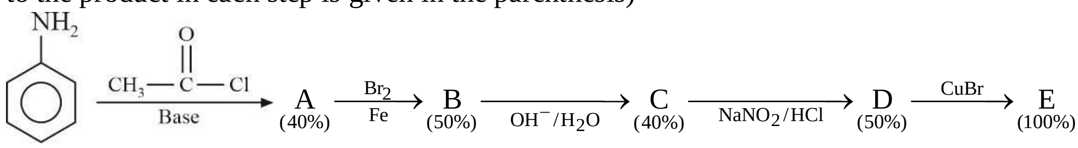
**9.** The percentage of available chlorine in a sample of 3.55 gram of bleaching powder which was dissolved in 100 ml of water is x%. 25 ml of this solution, on treatment with KI and dilute acid, required 20 ml of 0.125 N sodium thiosulphate solution. The value of 11x is _______.

**10.** The number of stereoisomers possible for the following compound are x.



The value of 16x is :

**11.** In the following reaction sequence the amount of E (in gram) formed from 25 moles of aniline is _____. (Atomic weight in g/mol; H = 1, C = 12, Br = 80, N = 14, O = 16. The yield (%) orresponding to the product in each step is given in the parenthesis)



**12.** NO and $O_2$ react to form $NO_2$ at 298 K according to the reaction.

$$NO_{(g)} + \frac{1}{2}O_{2(g)} \rightleftharpoons NO_{2(g)}$$

Initially pressure of NO and $O_2$ are 1 atm and 11/4 atm respectively and the final pressure of NO is $x \times 10^{-8}$ atm. The value of x is _______. [Use $(10^{6.13} = 1.34 \times 10^6)$]

(Given : $\Delta G_f^\circ (NO_2)_{(g)} = 52$ kJ/mol, $\Delta G_f^\circ (NO)_{(g)} = 87$ kJ/mol)

**13.** For a first order reversible reaction $A \underset{k_b}{\overset{k_f}{\rightleftharpoons}} B$, the initial concentration of A and B are $[A]_0$ and zero respectively. If concentrations at equilibrium are $[A]_{eq}$ and $[B]_{eq}$. The time taken to attain concentration of B to $\frac{[B]_{eq}}{2}$ is $\frac{1}{6}\ln 2$. $k_f$ and $k_b$ are rate constant of forward and backward reaction. $k_b - k_f = 2\ \text{min}^{-1}$ at 300 K. The value of $\Delta G^\circ$ in J/mol is _______.(Nearest Integer)

(Given: RT = 2500 at 300 K, $\log_{10} 2 = 0.3010$)

**14.** Consider an hypothetical electrochemical cell :

$$A_{(s)} | A^{2+}_{(aq)} || B^{2+}_{(aq)} | B_{(s)}$$

The standard emf of the given cell at 25°C and 20°C are 0.3525 V and 0.3533 V respectively. The value of magnitude $\Delta S^\circ$ for the cell in (J/K) is _______.

## SECTION 3

## MATRIX MATCH TYPE

This section contains **FOUR (04)** questions. Each question has **TWO (02)** matching lists: **LIST-I** and **LIST-II**. **FOUR** options are given representing matching of elements from **LIST-I** and **LIST-II**. **ONLY ONE** of these four options corresponds to a correct matching. For each question, choose the option corresponding to the correct matching.

**15.** Match each set of hybrid orbitals from List-I with complex(es) given in List-II.

| **List-I** | | **List-II** | |
|---|---|---|---|
| **(P)** | $d^2sp^3$ | **(1)** | $Ni(CO)_4$ |
| **(Q)** | $sp^3$ | **(2)** | $[Fe(en)_3]^{3+}$ |
| **(R)** | $sp^3d^2$ | **(3)** | $Na_2[Zn(CN)_4]$ |
| **(S)** | $dsp^2$ | **(4)** | $[NiCl_6]^{2-}$ |
| | | **(5)** | $[Co(C_2O_4)_3]^{3-}$ |
| | | **(6)** | $K_2[PtCl_4]$ |

The correct option is :

**(A)** (P)-(2), (Q)-(3), (R)-(4, 5), (S)-(1, 6)

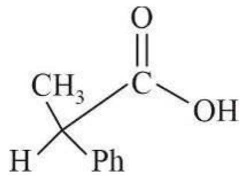
**(B)** (P)-(2, 5), (Q)-(1, 3), (R)-(4), (S)-(6)

**(C)** (P)-(2, 5), (Q)-(3), (R)-(4), (S)-(1, 6)

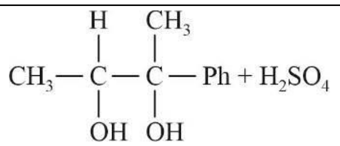
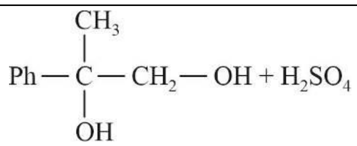
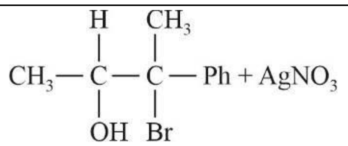
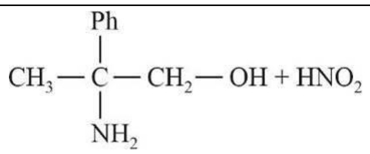
**(D)** (P)-(2), (Q)-(3, 6), (R)-(4, 5), (S)-(1)

**16.** The desired product X can be prepared by reacting the major product of the reactions in List-I with one or more appropriate reagents in List-II followed by acidification.

Given: order of migratory aptitude : (hydrogen > aryl > alkyl)



(X)

| **List-I** | | **List-II** | |
|---|---|---|---|
| **(P)** |  | **(1)** | $Br_2$, NaOH |
| **(Q)** |  | **(2)** | $[Ag(NH_3)_2]OH$ |
| **(R)** |  | **(3)** | HCHO, NaOH |
| **(S)** |  | **(4)** | Fehling solution |
| | | **(5)** | NaOCl |

The correct option is :

**(A)** (P)-(1, 5), (Q)-(2, 3, 4), (R)-(1, 5), (S)-(2, 3, 4)

**(B)** (P)-(1), (Q)-(2, 4), (R)-(1), (S)-(2, 4)

**(C)** (P)-(1, 5), (Q)-(2, 4), (R)-(1, 5), (S)-(2, 4)

**(D)** (P)-(1), (Q)-(2, 3, 4), (R)-(1), (S)-(2, 3, 4)

**17.** List-I contains reactions and List-II contains major products.

| | List-I | | List-II |
|---|---|---|---|
| **(P)** | $CH_3-C(CH_3)_2-ONa + CH_3-CH(Br)-CH_2-CH_3$ | **(1)** | $CH_3-CH(CH_3)-I$ |
| **(Q)** | $CH_3-CH_2-CH_2-CH_2-OH + H_2SO_4/\Delta$ | **(2)** | $CH_3-CH(CH_3)-OH$ |
| **(R)** | $CH_3-CH(CH_3)-O-CH_2-CH_3 + \underset{(1\text{ mol})}{HI}$ | **(3)** | $CH_3-CH_2-CH=CH_2$ |
| **(S)** | $CH_3CH(CH_3)-OH + HI$ | **(4)** | $CH_3-CH=CH-CH_3$ |
| | | **(5)** | $CH_3CH_2OH$ |

Match each reaction in List-I with one or more products in List-II and choose the correct option:

**(A)** (P)-(3), (Q)-(4), (R)-(2), (S)-(1)

**(B)** (P)-(4), (Q)-(3), (R)-(2), (S)-(1)

**(C)** (P)-(3), (Q)-(4), (R)-(5), (S)-(1)

**(D)** (P)-(4), (Q)-(3), (R)-(5), (S)-(1)

**18.** Dilution process of different aqueous solution, with water are given in List-I. The effects of dilution of the solution on $[H^+]$ are given in List-II. (Note : Degree of dissociation $(\alpha)$ of weak acid and weak base is << 1, degree of hydrolysis of salt << 1, $[H^+]$ represents the concentration of $H^+$ ions).

| | List-I | | List-II |
|---|---|---|---|
| **(P)** | (100 ml of 0.2 M $NH_4OH$ + 100 ml of 0.1 M HCl) diluted to 400 ml | **(1)** | The value of $[H^+]$ change to $\frac{1}{2}$ times of initial value on dilution |
| **(Q)** | (100 ml of 0.2 M $CH_3COOH$ + 100 ml of 0.2 M NaOH) diluted to 800 ml | **(2)** | The value of $[H^+]$ changes to $\frac{1}{3}$ times of initial value on dilution |
| **(R)** | (100 ml of 0.2 M $CH_3COOH$ + 100 ml of 0.4 M NaOH) diluted to 600 ml | **(3)** | The value of $[H^+]$ does not change on dilution |
| **(S)** | (100 ml of 0.1 M $NH_4OH$ + 100 ml of 0.4 M HCl) diluted to 600 ml | **(4)** | The value of $[H^+]$ changes to 3 times of initial value on dilution |
| | | **(5)** | The value of $[H^+]$ changes to 2 times of initial value on dilution |

Match each process given in List-I with one or more effect(s) in List-II. The correct option is :

**(A)** (P)-(1), (Q)-(5), (R)-(2), (S)-(4)

**(B)** (P)-(3), (Q)-(5), (R)-(2), (S)-(4)

**(C)** (P)-(3), (Q)-(5), (R)-(4), (S)-(2)

**(D)** (P)-(3), (Q)-(1), (R)-(4), (S)-(2)

## SUBJECT III : MATHEMATICS 60 MARKS

## SECTION 1

## MULTIPLE CORRECT ANSWERS TYPE

**This section consists of SIX (06) Multiple Correct Answers Type Questions**. Each question has 4 choices (A), (B), (C) and (D), out of which **ONE OR MORE THAN ONE CHOICE** is correct.

**1.** Let $f(n)=\sum_{r=1}^{n}\cot^{-1}\left(\frac{r^3}{2}+\frac{r}{2}+\frac{1}{r}\right)$; the inverse trigonometric functions assume values from their principal branch. Which of the following statement statement(s) is (are) true ?

**(A)** $\lim_{n\to\infty} f(n)=\frac{\pi}{4}$

**(B)** $\lim_{n\to\infty} f\left(\frac{1}{n}\right)-\lim_{n\to\infty} f(2n)=0$

**(C)** $\frac{\tan\left(f(n)+\frac{\pi}{4}\right)}{\cot\left(f(n)-\frac{\pi}{4}\right)}=1$

**(D)** $\lim_{n\to\infty}[f(n)]=0$; where $[x]$ represents greatest integer value of $x$

**2.** Each of circles $x^2+y^2+4y-1=0$ ; $x^2+y^2+6x+y+8=0$ and $x^2+y^2-4x-4y-37=0$ touch the other two then which of the following statement(s) is/are correct ?

**(A)** Point of concurrence of the tangents at point of contacts of circle is (–3, –3)

**(B)** Point concurrence of the tangents at point of contacts of circle is (–2, –1)

**(C)** Point of contacts of circles form the vertices of scalene triangle

**(D)** Centroid of triangle formed by centres of circles is $\left(-\frac{1}{3},-\frac{1}{6}\right)$

**3.** Let system of equations $-x+2y+5z=a$, $2x-4y+3z=b$, $x-2y+2z=c$.
Have atleast one solution for some non-zero real values of $(a, b, c)$ then which of the following system of equation(s) is/are inconsistent?

**(A)** $x+2y+z=a$, $2x+4y+2z=b$, $3x+6y+3z=c$

**(B)** $x+y+3z=a$, $5x+2y+6z=b$, $2x+y+3z+c=0$

**(C)** $x+y=a$, $y+z=b$, $z+x=c$

**(D)** $x+y+z=a$, $2x+2y+2z=-b$, $x+y+z=2c$

**4.** The normal at points $A$ and $B$ on a parabola $y^2=4ax$ intersect at a point $C$ on the curve. If $M$ is mid-point of $AB$, $N$ is the mid-point of $MC$ then which of the following statement(s) is/are correct?

**(A)** Locus of mid-point of $MC$ is a conic with latus rectum $a/3$

**(B)** Length of the common chord of locus of mid-point of $MC$ and given curve is $\frac{4a}{\sqrt{11}}$ units

**(C)** Eccentricity of locus of mid-point of $MC$ is 1

**(D)** Eccentricity of locus of mid-point of $MC$ is $\sqrt{2}$

**5.** Let $A(z_1),\ B(z_2),\ C(z_3)$ and $D(z_4)$ be the vertices of a trapezium in this order in anti-clokwise sense in argand plane let $|z_1 - z_2| = 4,\ |z_3 - z_4| = 10$ and diagonals $AC$ and $BD$ intersect at $P$.

It is also given that $\arg\left(\dfrac{z_4 - z_2}{z_3 - z_1}\right) = \dfrac{\pi}{2}$ and $\arg\left(\dfrac{z_3 - z_2}{z_4 - z_1}\right) = -\dfrac{\pi}{4}$ then which of the following statement(s) is/are correct ?

**(A)** Area of triangle $PCB$ is equal to $\dfrac{200}{21}$ square unit

**(B)** Area of trapezium $ABCD$ is equal to $\dfrac{140}{3}$ square unit

**(C)** $|CP - PD|$ is $\dfrac{10}{\sqrt{21}}$ unit

**(D)** Product of diagonals of trapezium is $\dfrac{100}{7}$

**6.** Let $f$ be a function differentiable throughout its domain such that $\lim\limits_{t\to x} \dfrac{e^t f(x) - e^x f(t)}{(t-x)\,(f(x))^2} = 3$ and $f(0) = 1$ then which of the following statement(s) is/are correct ?

**(A)** $f(0) + f'(0) = -1$

**(B)** $f'(0) = -2$

**(C)** $4e^{x-1} > 3x + 1$ in $x \in (1, 2)$

**(D)** $f(x)$ is increasing function for $\forall\ x < 0$

## SECTION 2

## NUMERICAL VALUE TYPE QUESTIONS

**This section consists of EIGHT (08) Numerical Value Type Questions**. The answer to each question is a **NUMERICAL VALUE**. For each question, enter the correct numerical value of the answer. If the answer is a decimal numerical value, then round-off the value to TWO decimal places. If the answer is an Integer value, then do not add zero in the decimal places. *In the OMR, do not bubble the ⊕ sign for positive values. However, for negative values, Θ sign should be bubbled*. (Example: 6, 81, 1.50, 3.25, 0.08).

**7.** The value of $\displaystyle\int_0^{\pi/4} \sqrt[4]{\text{cosec}^3 x \cdot \sec^5 x}\, dx$ is :

**8.** Let $F(x) = \begin{vmatrix} 1 + x\sin\alpha & \cos(x+\alpha) & \sin(x+\alpha) \\ 2 + x\sin\beta & \cos(x+\beta) & \sin(x+\beta) \\ 3 + x\sin\gamma & \cos(x+\gamma) & \sin(x+\gamma) \end{vmatrix}$, the total number of integers in the range of $F'(x)$.

**9.** Let set $A$ have 5 elements and set $B$ have 3 elements. If '$a$' be number of one-one functions from set $B$ to set $A$ and '$b$' be number of onto functions from set $A$ to set $B$. Find value of $\left(\dfrac{a \cdot b}{1000}\right)$.

**10.** Let $f : R - \{n\pi\} \to R,\ n \in \mathbb{Z}$ be a differentiable function with $f\left(\dfrac{\pi}{2}\right) = e - 1$. If $y = f(x)$ satisfies the differential equation $\dfrac{dx}{dy} = \dfrac{1}{(e^{\sin x} - y)\cdot \cot x}$, then the value of $\lim\limits_{x\to 0} f(x)$ is :

**11.** A function $f(x)$ satisfies the relation; $f(x+y)=f(x)+f(y)+xy(x+y)\ \forall\ x, y \in R$. If $f'(0)=-1$, then $\log_2(f'(3))$ is _______.

**12.** Let $P$ be a point whose image $Q$ in the plane $x - y + 3z + 3 = 0$ lies in the plane $x + y + z = 0$. If the line segment $PQ$ is bisected by $y$-axis. Then the value of $(PQ)^2$ is _______.

**13.** Consider the cube in the first octant with side OP, OQ and OR of length 1, along the $x$-axis, $y$-axis and $z$-axis respectively where $O(0, 0, 0)$ is the origin. Let $\hat{l}$ represent the unit vector along any given line and unit vectors along the diagonals are $d_1, d_2, d_3$ and $d_4$. $[(\hat{l}\cdot d_1)^2+(\hat{l}\cdot d_2)^2+(\hat{l}\cdot d_3)^2+(\hat{l}\cdot d_4)^2]$ is_______. (where $[\cdot]$ represents Greatest Integer Function).

**14.** Let $X=({}^{10}C_1)^2+4({}^{10}C_2)^2+9({}^{10}C_3)^2+\ldots\ldots 100({}^{10}C_{10})^2$ where ${}^{10}C_r\ r\in\{1, 2, \ldots\ldots 10\}$ represents binomial coefficient. Then the value of $\dfrac{X}{221000}$ is _______.

## SECTION 3

## MATRIX MATCH TYPE

This section contains **FOUR (04)** questions. Each question has **TWO (02)** matching lists: **LIST-I** and **LIST-II**. **FOUR** options are given representing matching of elements from **LIST-I** and **LIST-II**. **ONLY ONE** of these four options corresponds to a correct matching. For each question, choose the option corresponding to the correct matching.

**15.** Let $E_1=\left(x\in R, x\notin\{-1,1\} \text{ and } \dfrac{x^2}{x^2-1}\geq 0\right)$ and $E_2=\left\{x\in E_1, \log_e \tan^{-1}\left(\dfrac{x^2}{x^2-1}\right) \text{ is a real number}\right\}$

Let $f: E_1 \to R$ be the function defined by $f(x)=\tan^{-1}\left(\dfrac{x^2}{x^2-1}\right)$

and $g: E_2 \to R$ be the function defined by $g(x)=\log_e \tan^{-1}\left(\dfrac{x^2}{x^2-1}\right)$

| **List-I** | | **List-II** | |
|---|---|---|---|
| **(P)** | The range of $f$ is | **(1)** | $(0, \infty)$ |
| **(Q)** | The range of $g$ contains | **(2)** | $(1, \infty)$ |
| **(R)** | The domain of $f$ contains | **(3)** | $(-\infty,-1)\cup(1,\infty)$ |
| **(S)** | The domain of $g$ is | **(4)** | $\{0\}$ |
| | | **(5)** | $\{0\}\cup(\pi/4,\ \pi/2)$ |

The correct option is:

**(A)** (P)-(5), (Q)-(4, 5), (R)-(2, 3), (S)-(3)
**(B)** (P)-(5), (Q)-(4), (R)-(2, 3, 4), (S)-(3)
**(C)** (P)-(3), (Q)-(5), (R)-(2, 3, 4), (S)-(1, 2, 3)
**(D)** (P)-(3), (Q)-(4), (R)-(2, 3), (S)-(3)

**16.** In a class, *a* team has to be formed from a group of 8 boys $B_1, B_2, B_3, B_4, B_5, B_6, B_7, B_8$ and 6 girls $G_1, G_2, G_3, G_4, G_5, G_6$.

**(i)** Let $N_1$ be the number of ways of forming a team such that the team has 6 members having equal number of boys and girls.

**(ii)** Let $N_2$ be the number of ways of forming a team such that the team has 5 members having 2 boys and 3 girls.

**(iii)** Let $N_3$ be the number of ways of forming a team such that the team has 10 members having atleast 7 boys and atleast 2 girls.

**(iv)** Let $N_4$ be the number of ways of forming a team such that the team has 5 members having atleast 3 boys and such that both $G_2$ and $B_3$ are NOT in the team.

| **List-I** | | **List-II** | |
|---|---|---|---|
| **(P)** | The value of $N_1$ is | **(1)** | 200 |
| **(Q)** | The value of $N_2$ is | **(2)** | 560 |
| **(R)** | The value of $N_3$ is | **(3)** | 1120 |
| **(S)** | The value of $N_4$ | **(4)** | 546 |
| | | **(5)** | 175 |
| | | **(6)** | 350 |

The correct option is:

**(A)** (P)-(3), (Q)-(2), (R)-(5), (S)-(4)
**(B)** (P)-(3), (Q)-(2), (R)-(6), (S)-(4)
**(C)** (P)-(3), (Q)-(1), (R)-(5), (S)-(6)
**(D)** (P)-(2), (Q)-(3), (R)-(6), (S)-(5)

**17.** Let $E : \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ when $b > a > 0$, be an ellipse in the *xy*-plane whose minor axis LM subtends an angle of 120° at one of its foci (Say *N*). Let the area of the triangle LMN be $\sqrt{3}$.

| **List-I** | | **List-II** | |
|---|---|---|---|
| **(P)** | The length of minor axis of *E* is | **(1)** | 4 |
| **(Q)** | The eccentricity of *E* is | **(2)** | $2\sqrt{3}$ |
| **(R)** | The distance between the foci of *E* is | **(3)** | $\frac{1}{\sqrt{2}}$ |
| **(S)** | The length of latus rectum of *E* is | **(4)** | $\frac{1}{2}$ |
| | | **(5)** | 3 |
| | | **(6)** | 2 |

The correct options is :

**(A)** (P)-(1), (Q)-(4), (R)-(6), (S)-(5)
**(B)** (P)-(1), (Q)-(3), (R)-(6), (S)-(1)
**(C)** (P)-(2), (Q)-(4), (R)-(6), (S)-(5)
**(D)** (P)-(2), (Q)-(3), (R)-(5), (S)-(5)

**18.** Let $f_1 : R \to \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$, $f_2 : R \to R$, $f_3 : R - \{0\} \to [-1, 1] - \{0\}$, $f_4 : R \to R$.

**(i)** $f_1(x) = \sin^{-1}\left(\frac{2x}{1+x^2}\right)$

**(ii)** $f_2(x) = \sqrt[3]{x^2 |x|} + |x| - 1$

**(iii)** $f_3(x) = \begin{cases} \frac{x}{2x^2 + |x|} & , \quad x \neq 0 \\ 1 & , \quad x = 0 \end{cases}$

**(iv)** $f_4(x) = \text{sgn}(x) + \text{sgn}(-x)$ (where sgn($x$) represents signum function)

| List-I | | List-II | |
|---|---|---|---|
| **(P)** | The function $f_1$ is | **(1)** | NOT continuous at $x = 0$ |
| **(Q)** | The function $f_2$ is | **(2)** | Continuous at $x = 0$ and NOT differentiable at $x = 0$ |
| **(R)** | The function $f_3$ is | **(3)** | Differentiable at $x = 0$ and its derivative is NOT continuous at $x = 0$ |
| **(S)** | The function $f_4$ is | **(4)** | Differentiable at $x = 0$ and its derivative is continuous at $x = 0$ |

**(A)** (P)-(2), (Q)-(3), (R)-(1), (S)-(4)
**(B)** (P)-(4), (Q)-(2), (R)-(1), (S)-(4)
**(C)** (P)-(4), (Q)-(1), (R)-(2), (S)-(3)
**(D)** (P)-(1), (Q)-(2), (R)-(3), (S)-(4)

**End of JEE Advanced-11 | PAPER-2 | JEE-2022**